Analytical Cume
September 3, 2004

Equations: $q = it$
$q = nF$, Where F is 96,483 coul/equivalent

1. (50 points)
Chlorine has been used for decades to disinfect drinking water. An undesirable side effect of this treatment is the reaction of chlorine with organic impurities to create organochlorine compounds (RX), some of which could be toxic. Monitoring total organic halide is now required for many water providers.
A standard procedure for total organic halides is to pass water through activated charcoal that adsorbs organic compounds. Then the charcoal is combusted to liberate hydrogen halides (HX).

$$RX + O_2 \longrightarrow CO_2 + H_2O + HX$$

The HX is dissolved into aqueous solution and measured by automatic coulometric titration with a silver anode. The half reaction is:

$$X^-(aq) + Ag(s) \longrightarrow AgX(s) + e^-$$

When 1.0 L of drinking water was analyzed, a current of 4.23 mA was required for 387 s to consume all the X^-. A blank prepared by oxidizing charcoal required 6 s at 4.23 mA.
Express the total organic halide concentration in the drinking water sample as μM halogen.

2. (50 points)
$H_2S(aq)$ can be analyzed by titration with coulometrically generated I_2.

$$H_2S + I_2 \longrightarrow S(s) + 2H^+ + 2I^-$$

To 50.00 mL of sample were added 4 g of KI. The electrolysis required 812 s at 52.6 mA. Calculate the concentration (μg/mL) of H_2S (molar mass 34g/mol) in the sample.
Electrochemical generation reaction is:

$$I_2 + 2e^- \rightarrow 2I^-$$